

Definitive Treatment of Oral Infections

Definitive treatment requires either **extraction** or **root canal therapy**. If **periodontitis** is the etiology, **antibiotics** and **periodontal surgery** are indicated. Rapid resolution occurs with removal of the source of infection.

Actinomycosis

Actinomyces (see Chap. 185) are filamentous bacteria commonly found in the oral cavity. *Actinomyces israelii*, a pathogenic strain, causes **actinomycosis**, particularly in young adults—more commonly men—following oral surgery or trauma.

Lumpy jaw, the most common form of actinomycosis, presents with indurated, violaceous cervical and submandibular masses due to abscesses and sinus tracts. Purulent drainage containing **small yellow sulfur granules** is characteristic. Diagnosis is confirmed by: - Positive anaerobic culture - Microscopic examination of sulfur granules - Fluorescent antibody testing

Candidiasis

Candida albicans (see Chap. 189) is a commensal yeast normally present in the gastrointestinal tract, vagina, and oral cavity. It becomes pathogenic in compromised oral environments, such as: - Under dental appliances - Xerostomia - HIV infection - Diabetes mellitus - Use of systemic antibiotics, chemotherapy, or immunosuppressant medications

Diagnosis is suggested by clinical presentation, positive **potassium hydroxide (KOH) examination**, and response to antifungal therapy. When diagnosis is uncertain, cultures for *Candida* species may be performed—though they can be non-diagnostic, as 20% to 60% of healthy individuals carry *Candida* as part of normal oral flora.

Clinical Forms of Oral Candidiasis

- **Pseudomembranous candidiasis (thrush):**

Most common in infants, individuals on antibiotics or corticosteroids, and immunocompromised patients. Presents as curd-like white plaques that wipe off, revealing an underlying erythematous but intact mucosa. Symptoms include burning mouth, dryness, and thickened tongue and lips.

- **Erythematous candidiasis:**

Characterized by localized or generalized erythema, mucosal atrophy, and burning sensation. Often a complication of broad-spectrum antibiotic use. Atrophy of filiform papillae is common.

- **Chronic atrophic candidiasis (denture stomatitis):**

Asymptomatic, well-demarcated erythematous patches beneath dentures, especially when worn during sleep and with poor oral hygiene.

- **Median rhomboid glossitis:**

A diamond- or oval-shaped, erythematous, atrophic, and depapillated lesion on the midline of the dorsal tongue. Frequently seen in patients with **acquired immunodeficiency syndrome (AIDS)**.

- **Angular cheilitis:**

Discussed separately in the section on Angular Cheilitis (Perlèche).

- **Hyperplastic candidiasis:**

An uncommon form presenting as a thickened, firm, white plaque, often in smokers at sites of friction (e.g., anterior buccal mucosa). Lesions are fixed and do not rub off.

- **Chronic mucocutaneous candidiasis:**

May be localized or diffuse. Can be associated with endocrinopathies and thymoma (see Chap. 144).

Opportunistic Fungal Infections

A wide range of viruses and fungi can cause disease in immunosuppressed individuals. Oral involvement may result from primary infection or hematogenous dissemination.

Common Pathogens:

- **Viruses:** Herpes simplex virus (HSV), cytomegalovirus (CMV), human papillomavirus (HPV)
- **Fungi:** *Aspergillus*, *Zygomycetes*, *Fusarium*, *Trichosporon*, *Histoplasma capsulatum*, *Blastomyces dermatitidis*, *Cryptococcus neoformans*, *Coccidioides immitis*, *Paracoccidioides brasiliensis*

Clinical Presentation:

- **Viral infections:** Typically present as erosions, ulcerations, or papillomas.
- **Fungal infections:** Usually manifest as chronic, verrucous, or necrotic mucosal ulcerations.

Diagnosis and Management:

- **Biopsy and culture** are essential for definitive diagnosis.
- **Aggressive treatment** with appropriate antiviral or antifungal agents is required.

Key References

1. Tyring S: *Mucocutaneous Manifestations of Viral Diseases*. New York, Marcel Dekker, Inc, 2002
2. Eisen D, Lynch D: *The Mouth: Diagnosis and Treatment*. St. Louis, Mosby-Year Book, Inc, 1998

The full reference list for all chapters is available at www.digim7.com.